Introduction

SWAT+ Documentation

docs.swat.tamu.edu

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SWAT+ is a watershed model. It simulates water, sediment, nutrient, pesticide, and crop yield response to climate and management at scales from a single field up to a large river basin.

This section covers what the model is, how it differs from the original SWAT, how to cite it, and how releases are numbered.

Pages

- What is SWAT+
- SWAT vs SWAT+
- Citing and license
- Release notes

The Soil and Water Assessment Tool Plus (SWAT+) is an open-source watershed model developed jointly by the USDA Agricultural Research Service (USDA-ARS) and Texas A&M AgriLife Research, with contributions from Colorado State University, the USDA-ARS Hydrology and Remote Sensing Laboratory and other institutions.

SWAT+ simulates the quantity and quality of surface water and groundwater, from small watersheds to large river basins. It is used to evaluate the effects of land use, land management, soils and climate change on water resources.

Background

Over more than two decades the original Soil and Water Assessment Tool (SWAT) became one of the most widely used watershed models in the world. Its many applications showed limits in how a watershed could be represented,

and the additions made over the years left a code base that was hard to maintain and extend. SWAT+ is a restructured version of SWAT. Most of the process equations are the same, but the code is organized around objects and the input files are relational, which makes the model easier to maintain and lets new science be added as separate modules. SWAT+ also allows a more flexible spatial representation: landscape units, HRUs, aquifers, channels and reservoirs can be connected in any arrangement through connection files. The main differences are listed on SWAT vs SWAT+.

Processes simulated

For each HRU, channel, aquifer, reservoir and wetland, SWAT+ keeps a daily mass balance of:

- water: precipitation, snow, evapotranspiration, surface runoff, lateral flow, percolation, groundwater flow and channel routing
- sediment: detachment, transport and deposition on the landscape and in channels
- nutrients: nitrogen and phosphorus cycling in the soil, plant uptake, transport with runoff and subsurface flow, and in-stream transformations
- carbon: soil organic carbon and residue pools
- pesticides and pathogens: application, decay and transport
- salts and other constituents
- plant growth: biomass, leaf area, yield, and water, nutrient and temperature stress

Management (planting, tillage, fertilizer and manure, irrigation, harvest, grazing) can follow fixed schedules or rules in decision tables. Water can be allocated between sources and users, and structural practices such as tile drains, filter strips and terraces can be represented.

Output can be written daily, monthly, yearly and as average annual values. The Model Theory section describes each process.

Applications

- effects of management practices on water yield and water quality
- soil erosion and non-point source pollution
- crop yield response to climate and management
- watershed planning and total maximum daily load (TMDL) studies
- climate change impacts on water resources
- irrigation and water allocation

Running the model

SWAT+ is a command-line program. It reads a folder of plain-text input files that describe the watershed, climate, soils, land use, management and simulation period, and writes plain-text (and optionally CSV) output files to the same folder.

Most users do not write the input files by hand. They use the supporting tools:

- **QSWAT+**, a QGIS plugin, delineates the watershed from a DEM, creates HRUs from soil, land use and slope maps, and writes the project database.
- **SWAT+ Editor** edits the inputs of a project, writes the input files and runs the model.
- **SWAT+ Toolbox** runs sensitivity analysis, calibration and other tasks on a finished project.

All three can be installed from install.swat.tamu.edu. See Install.

Source and license

The source code is at github.com/swat-model/swatplus and is released under the GNU Lesser General Public License 2.1. See Citing and license.

Sources

- README.md and CITATION.cff in the source repository at 62.0.1.
- Input file groups: `src/input_file_module.f90`.

SWAT+ is the restructured successor to SWAT (Soil and Water Assessment Tool). Most of the process equations are the same. The code structure, the input files and the way parts of a watershed are connected are different. The SWAT column below refers to SWAT2012, the last release of the original model (swat-model/swat2012).

Spatial structure

	SWAT2012	SWAT+
Units	subbasins, each with HRUs and one main channel	landscape units, routing units, HRUs, channels, aquifers, reservoirs, wetlands, point sources, each a separate object
Connections	fixed: HRUs drain to their subbasin's channel, subbasins drain downstream in numbered order	defined in connection files (`*.con`): any object can send any part of its flow (surface runoff, lateral flow, tile flow, recharge) to any other object
Floodplain and upland	not separated	landscape units can separate upland and floodplain areas

The connection files make it possible, for example, to route upland runoff across a floodplain before it reaches the channel, to send tile flow from one HRU to a wetland, or to divide a reservoir's outflow between two channels.

Input files

SWAT2012 used one set of files per subbasin and HRU (`.sub`, `.hru`, `.mgt`, `.gw`, `.rte`, `.sol`, …), each repeating the full parameter list.

SWAT+ uses relational input files grouped by purpose. A parameter set is written once, given a name, and referenced by name from every object that uses it. For example, one row of `hydrology.hyd` can serve hundreds of HRUs. The master file `file.cio` lists every input file by group:

- simulation: `time.sim`, `print.prt`, `object.cnt`
- basin: `codes.bsn`, `parameters.bsn`, `carbon.bsn`
- climate: `weather-sta.cli`, `weather-wgn.cli`, `pcp.cli`, `tmp.cli` and the station files
- connections: `hru.con`, `rout_unit.con`, `aquifer.con`, `chandeg.con`, `reservoir.con`, `recall.con`, …
- HRUs, land use and management: `hru-data.hru`, `landuse.lum`, `management.sch`, `cntable.lum`, decision tables (`*.dtl`)
- hydrology: `hydrology.hyd`, `topography.hyd`, `field.fld`
- soils and plants: `soils.sol`, `nutrients.sol`, `plants.plt`, `plant.ini`
- databases: fertilizer, tillage, pesticide, urban and other parameter tables

The Input Reference describes each file.

Management

SWAT2012 scheduled operations in the `.mgt` file of each HRU, by date or by fraction of heat units.

SWAT+ management schedules (`management.sch`) are shared between HRUs, and can combine dated operations with decision tables (`*.dtl`). A decision table checks conditions every day (soil water, plant growth stage, date, crop rotation year, reservoir level, and so on) and applies an action when they are met. Decision tables also drive reservoir releases, flow control and land use change. See Decision tables.

Groundwater

SWAT2012 had a shallow and a deep aquifer for each HRU, with the shallow aquifer feeding the main channel of its subbasin.

SWAT+ aquifers are separate objects, usually one per landscape unit or group of HRUs, connected to channels through `aquifer.con`. SWAT+ also includes gwflow, a gridded groundwater module that simulates water table heads and exchanges with channels, soils, tile drains, reservoirs and wells cell by cell. See Aquifers.

Additional processes

SWAT+ adds or extends: a carbon module with soil organic matter pools, water allocation between sources and users, salts and other constituents, rice paddies, and a sediment budget for channels. Each is described in the Model Theory section.

Outputs

SWAT2012 wrote a fixed set of files (`output.hru`, `output.sub`, `output.rch`, `output.std`).

SWAT+ writes one file per object type, process and time step, switched on and off in `print.prt`. Examples from the `Ames_sub1` reference dataset:

```
basin_wb_aa.txt        basin water balance, average annual
hru_wb_aa.txt          HRU water balance, average annual
channel_sd_day.txt     channel flow and loads, daily
basin_aqu_aa.txt       aquifer, average annual
```

The suffix gives the time step: `_day`, `_mon`, `_yr` or `_aa`. Each row of `print.prt` switches the four time steps on or off. See the Output Reference.

Code

	SWAT2012	SWAT+
Language	Fortran, fixed form	Fortran, free form
Organization	shared common blocks	modules grouped by process and object
Source	released as archives	public on GitHub, LGPL-2.1
Build	hand-written makefiles	CMake

Migration

There is no automatic converter from a SWAT2012 project to SWAT+. Rebuild the project in QSWAT+ from the same DEM, soil, land use and climate data, then transfer calibrated parameter values by hand, checking each against the SWAT+ units and meaning.

Sources

- Connection and input file names: `type input_con` and the other `input_*` types in `src/input_file_module.f90`.
- Aquifer objects: `src/aquifer_module.f90`, read by `aqu_read.f90`. Gridded groundwater: `gwflow_*.f90`. Spatial aquifer connections (`aquifer2d.con`): `aqu2d_read.f90`, `aqu2d_init.f90`.
- Build: `CMakeLists.txt` compiles `src/*.f90` free form (`-free` for Intel, `-ffree-line-length-none` for GNU) into one executable.
- SWAT2012: `swat-model/swat2012`.

License

SWAT+ is released under the GNU Lesser General Public License, version 2.1 (LGPL-2.1). The full text is in the `LICENSE` file of the source repository.

Citing the software

Cite the model release used in the study, with its version and DOI. The author list below is the one in the repository's citation file at 62.0.1:

> Arnold, J. G., Čerkasova, N., White, M. J., Bailey, R., Thorp, K., Jeong, J., Zhang, X., Ugraskan, T., van Griensven, A., Rathjens, H., Raj, C., Cai, X., Geter, W. F., David, O., Carlson, J. R., and Chawanda, C. J. Soil and Water Assessment Tool Plus (SWAT+), version 62.0.1. https://doi.org/10.5281/zenodo.14983533

Releases are archived on Zenodo:

DOI	Points to
10.5281/zenodo.14983533	all versions; resolves to the newest archived release
10.5281/zenodo.22919722	release 62.0.0
10.5281/zenodo.14983534	release 61.0.2

When a release has a version DOI, cite that DOI so the exact code can be found again.

GitHub shows a "Cite this repository" button on the repository page, built from the same citation file, with APA and BibTeX forms.

Citing the documentation

When a statement comes from this site, cite the page and the documentation version shown in the version selector at the top of the page, for example "SWAT+ Documentation, revision 62.0.1, Soft calibration, https://docs.swat.tamu.edu/62.0.1/calibration/soft-calibration/".

Older versions

Revisions before the move to GitHub are in the modular_swatplus repository on Bitbucket.

Sources

- `CITATION.cff`, `LICENSE` and `README.md` at 62.0.1.
- Zenodo records 14983533 (concept), 22919722 and 14983534, checked 20 October 2026.

SWAT+ releases are listed on the GitHub releases page, with prebuilt executables and the merged pull requests of each release. Version numbers are explained in Tagging and releases.

This page lists the changes in each release that affect model users: input files, outputs and results. Each version of this documentation describes one release; use the version selector at the top of the page for earlier ones.

62.0.1

Released 19 October 2026.

Input files

- `file.cio`, `water_rights` line: now names two files, the places of use (`place_of_use.wro`) and the points of diversion (`point_of_diver.wro`), in place of the three water rights files. The water allocation input files were reorganized around places of use, points of diversion and return; see Water allocation. A `file.cio` written for 62.0.0 or earlier must be updated; see Older file.cio files.
- `parameters.bsn`: two columns that were unused are now read as the pesticide groundwater mixing factor and a switch for temperature-dependent pesticide decay. SWAT+ Editor still labels them `lin_sed` and `exp_sed`; a value of 1 in the second turns temperature-dependent decay on. See parameters.bsn.
- `pesticide.pes`: three new columns for plant uptake. Files in the older layout are still read, with no plant uptake. See pesticide.pes.

- Recall: the recall file named in `file.cio` is read (instead of a fixed name), and empty recall files are no longer created. See recall.
- `exco.exc` and `exco_om.exc`: the original layout is restored.
- Manure: a manure database (`manure_db.frt`, `manure_om.frt`) and manure allocation inputs were added. See manure allocation.
- Transplanting: transplant data are read from `transplant.ops`; `transplant.plt` is no longer read.

Outputs

- Water allocation outputs were replaced by place-of-use based files.
- New sediment budget outputs for channels, stream orders and the basin, with corrected average annual values.
- Rice paddy output, and paddy water included in the ET reports.
- gwflow outputs now follow their `print.prt` and gwflow switches, and the average annual storage change is computed correctly.
- The legacy carbon output no longer writes duplicate rows for each HRU.

See the Output Reference for the files.

Model changes

- Rice paddies: seepage, weir spill volume and the impoundment switch day were revised, and soil evaporation is skipped while paddy water remains.
- Canopy interception is applied for all runoff methods.
- Pesticides: temperature-dependent decay, plant uptake and mass balance corrections.
- Water allocation and irrigation were reworked; the irrigation days of an unlimited source are reset correctly.

Calibration

- `aquifer_delay` (gwflow) is changed in days.
- New object types `gwf_riv` and `gwf_sgl` for gwflow channel and single-value parameters.
- `nperco_lchtile` changes the right parameter.

See Parameter change files.

Fixes

- A crash in tillage mixing for plant communities with more than 12 plants.
- An endless loop when all operations of a management schedule share one date.
- Root depth of perennials stuck at 25.4 mm at a cold start.
- Release executables built with gfortran no longer stop with a floating-point trap on macOS, and the macOS executable no longer needs Homebrew's gcc libraries.

Documentation for 62.0.1

This version of the documentation was rewritten against the 62.0.1 source. Pages now describe what the model does, including inputs that are read but have no effect and behaviour that differs from the input file labels, so that results can be interpreted correctly. Examples:

- Measured PET is replaced by the Hargreaves estimate within the same day, so `pet` 3 in `codes.bsn` gives Hargreaves results (codes.bsn).
- Name lookups between files fall back to the first entry without a message when a name does not match; check `diagnostics.out` and the names themselves (weather-sta.cli, hru-data.hru).
- Several basin parameters cannot be changed through `calibration.cal` (Parameter change files).
- Manure allocation, street sweeping and the pathogen simulation have no effect, and water allocation behaves as its inputs describe only for a single place of use (Water allocation).

Each page states these points where they apply.

Sources

- SWAT+ 62.0.1 release and `git log 62.0.0..62.0.1` in the source repository.
- Input changes: `type input_water_allocation` in `src/input_file_module.f90` and `read (107,*) name, in_wallo` in `src/readcio_read.f90`; `pestgwfact` and `temp_decay` replacing `spcon` and `spexp` in `src/basin_module.f90`; `src/pesticide_data_module.f90`; `src/recall_read.f90` and `src/recalldb_read.f90`; `src/exco_db_read.f90` and `src/exco_read_om.f90`; `src/plant_transplant_read.f90`.
- Outputs: `src/wallo_pou_output.f90`, `src/basin_sedbud_output.f90`, `src/ch_sedbud_output.f90`, `src/chord_sedbud_output.f90`, `src/gwflow_output.f90`.
- Build: `CMakeLists.txt` (gfortran flags and macOS linking).